Gavin's Lectures on Physics - Particle Physics



Author：长颈鹿 Gavin

July 24, 2026

Contents

1 Fundamental Particle Properties **4**
- 1.1 Quarks . . . 4
- 1.2 Leptons . . . 4
- 1.3 Gauge Bosons . . . 4
- 1.4 Higgs Boson . . . 4
- 1.5 Summary of Generations . . . 5

2 Fundamental Quantum Numbers and the Quark Model **6**
- 2.1 Gell-Mann–Nishijima Charge Formula . . . 6
  - 2.1.1 General form (all 6 quark flavors) . . . 6
  - 2.1.2 Restricted 3-quark approximation (u, d, s only) . . . 6
  - 2.1.3 Proof via fundamental quark charges . . . 6
- 2.2 Definitions of Core Quantum Numbers . . . 6
- 2.3 Isospin Values for Quarks . . . 7
- 2.4 Worked Examples . . . 7
  - 2.4.1 Example 1: Ω^- (sss) . . . 7
  - 2.4.2 Example 2: Δ^{++} (uuu) . . . 7

3 Relativistic Field Equations (by Spin) **8**
- 3.1 Spin-0: Klein–Gordon Equation (Scalar Bosons) . . . 8
- 3.2 Spin-1/2: Dirac Equation (Fermionic Spinors) . . . 8
  - 3.2.1 Time-dependent form . . . 8
  - 3.2.2 Covariant form (natural units) . . . 8
  - 3.2.3 Standard Dirac matrix representation . . . 8
  - 3.2.4 Pauli matrices . . . 8
  - 3.2.5 Chiral structure . . . 8
  - 3.2.6 Massless Weyl fermions . . . 9
- 3.3 Spin-1: Vector Field Equations . . . 9
  - 3.3.1 Abelian (massless photon, QED): Maxwell equations . . . 9
  - 3.3.2 Non-Abelian (gluons, W/Z bosons): Yang–Mills equations . . . 9
- 3.4 Spin-3/2: Rarita–Schwinger Equation . . . 9
- 3.5 Spin-2: Vacuum Einstein Field Equation (Gravity) . . . 9

4 Perturbative Calculation Framework **10**
- 4.1 Fermi's Golden Rule . . . 10
- 4.2 Feynman Path Integral Formalism . . . 10
- 4.3 Feynman Rules . . . 10
  - 4.3.1 Position-space ϕ^4 Theory . . . 10
  - 4.3.2 Momentum-space General Rules . . . 11
  - 4.3.3 External Line Factors (asymptotic states) . . . 11
  - 4.3.4 Internal Line Propagators (virtual particles) . . . 11
  - 4.3.5 Vertex Factors . . . 12
  - 4.3.6 Supplementary Rules . . . 12
  - 4.3.7 Example: Muon-electron scattering . . . 12

5 Standard Model Gauge Group Structure **13**
5.1 Conservation Laws Summary for All Interactions . . . 13
5.2 Full Standard Model Gauge Symmetry . . . 13
5.3 $SU(3)_c$ – Strong Interaction (QCD) . . . 13
5.4 $SU(2)_L$ – Weak Interaction (Left-Handed Chiral Sector) . . . 14
5.5 $U(1)_Y$ – Hypercharge Gauge Symmetry . . . 14
5.6 Electroweak Symmetry Breaking . . . 15

6 Full Standard Model Lagrangian **16**
6.1 Gauge Field (Yang–Mills) Sector . . . 16
6.2 Fermion Kinetic (Matter) Sector . . . 16
6.3 Higgs Sector . . . 16
6.4 Yukawa Coupling (Fermion Mass Generation) . . . 17

7 Quantum Electrodynamics (QED) – Electromagnetic Interaction **18**
7.1 QED Action and Lagrangian Density . . . 18
7.1.1 Component Lagrangians . . . 18
7.1.2 Full QED Lagrangian . . . 18
7.2 Gauge Invariance . . . 18
7.3 QED Vertex and Feynman Rules . . . 19

8 Quantum Chromodynamics (QCD) – Strong Interaction **20**
8.1 Color Charge and Gluon States . . . 20
8.1.1 8 physical gluon octet states . . . 20
8.1.2 1 unobservable color singlet . . . 20
8.2 Color Confinement . . . 20
8.3 Asymptotic Freedom . . . 20
8.3.1 QCD static quark potential . . . 21
8.4 QCD Vertex and Feynman Rules . . . 21
8.5 Experimental Evidence: R Ratio in e^+e^- Annihilation . . . 21
8.6 Quark Model: Baryon Flavor Wavefunctions . . . 22
8.6.1 $SU(3)_f$ Three-Quark Decomposition . . . 22
8.6.2 Fully Symmetric Decuplet ($\mathbf{10}$) . . . 22
8.6.3 Fully Antisymmetric Singlet ($\mathbf{1}$) . . . 22
8.6.4 Mixed-Symmetry Octets ($\mathbf{8}, \mathbf{8}$) . . . 22
8.6.5 Clebsch–Gordan Coefficients . . . 23

9 Weak Interaction **24**
9.1 Charged Current Interaction . . . 24
9.2 Neutral Current Interaction . . . 24
9.3 Weak Interaction Vertex Factors . . . 24
9.4 Flavor Mixing in Weak Interactions . . . 25
9.4.1 Quark Mixing: CKM Matrix . . . 25
9.4.2 Lepton Mixing: PMNS Matrix . . . 25
9.5 Neutrino Physics . . . 26
9.5.1 Helicity . . . 26
9.5.2 Neutrino Detection . . . 26
9.5.3 Neutrino Sources . . . 26
9.5.4 Neutrino Oscillations . . . 26

10 Discrete Symmetries and CP Violation **27**
10.1 Charge Conjugation Operator $\hat{C}$ . . . 27
10.2 Parity Operator $\hat{P}$ . . . 27
10.2.1 Intrinsic parity assignments . . . 27
10.2.2 Hadron parity . . . 27
10.3 Time Reversal Operator $\hat{T}$ . . . 28
10.4 CPT Theorem . . . 28
10.5 CP Violation in the Neutral Kaon System . . . 28
10.5.1 Flavor eigenstates . . . 28
10.5.2 Pure CP eigenstates (theoretical) . . . 28
10.5.3 Physical mass eigenstates . . . 28

11 Beyond the Standard Model **29**
11.1 Fundamental Coupling Constants and Interaction Ranges . . . 29
11.2 Limitations of the Standard Model . . . 29
11.2.1 Anomalous magnetic moment . . . 29
11.3 Gauge Coupling Unification . . . 29
11.4 Dark Matter: WIMP Candidate . . . 30
11.5 Supersymmetry (SUSY) . . . 30
11.6 Seesaw Mechanism for Neutrino Masses . . . 30

A Appendix 1: Lie Group Notation Reference **31**

B Appendix 2: Matrix Definitions **31**

1 Fundamental Particle Properties

This section summarizes the key properties of all elementary particles in the Standard Model. Natural units $\hbar = c = 1$ are used; masses are given in GeV/c^2.

1.1 Quarks

Quarks are spin-1/2 fermions carrying color charge. They are confined into hadrons.

Quark	Flavor	Mass (GeV)	Q/e	I_3	B
u	up	~ 0.002	$+2/3$	$+1/2$	$1/3$
d	down	~ 0.005	$-1/3$	$-1/2$	$1/3$
s	strange	~ 0.095	$-1/3$	0	$1/3$
c	charm	~ 1.275	$+2/3$	0	$1/3$
b	bottom	~ 4.18	$-1/3$	0	$1/3$
t	top	~ 173.0	$+2/3$	0	$1/3$

Note: Quark masses are scheme-dependent and quoted at the $\overline{\text{MS}}$ scale; top quark mass is the pole mass.

1.2 Leptons

Leptons are spin-1/2 fermions that do not carry color charge. Neutrino masses are non-zero but extremely small.

Lepton	Mass (GeV)	Q/e	L	I_3 (L)
e	0.000511	-1	1	$-1/2$
μ	0.10566	-1	1	$-1/2$
τ	1.77686	-1	1	$-1/2$
ν_e	$< 10^{-9}$	0	1	$+1/2$
ν_μ	$< 10^{-9}$	0	1	$+1/2$
ν_τ	$< 10^{-9}$	0	1	$+1/2$

Note: Neutrino masses are not fixed by the Standard Model; the values shown are upper limits from oscillation experiments.

1.3 Gauge Bosons

Gauge bosons are spin-1 particles that mediate the fundamental forces.

Boson	Mass (GeV)	Q/e	Interaction	Spin
γ (photon)	0	0	Electromagnetic	1
g (gluon)	0	0	Strong	1
$W^\pm$	80.379	± 1	Weak (charged)	1
Z^0	91.1876	0	Weak (neutral)	1

1.4 Higgs Boson

The Higgs boson is a spin-0 scalar particle, responsible for spontaneous electroweak symmetry breaking and the generation of fermion masses.

Particle	Mass (GeV)	Q/e	I	J^P
H	125.10	0	0	0^+

1.5 Summary of Generations

The Standard Model contains three generations of fermions:

$$\begin{array}{ccc} \text{Generation I} & \text{Generation II} & \text{Generation III} \\ \begin{pmatrix} \nu_e \\ e \end{pmatrix} & \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix} & \begin{pmatrix} \nu_\tau \\ \tau \end{pmatrix} \\ \begin{pmatrix} u \\ d \end{pmatrix} & \begin{pmatrix} c \\ s \end{pmatrix} & \begin{pmatrix} t \\ b \end{pmatrix} \end{array} \tag{1}$$

2 Fundamental Quantum Numbers and the Quark Model

2.1 Gell-Mann–Nishijima Charge Formula

2.1.1 General form (all 6 quark flavors)

Relates electric charge, isospin, and additive flavor quantum numbers:

$$\frac{Q}{e} = I_3 + \frac{1}{2}Y \tag{2}$$

$$Y = B + S + C + \tilde{B} + T \tag{3}$$

$$\implies \frac{Q}{e} = \frac{1}{2}\left[B + \left(U + C + T + D + S + \tilde{B}\right)\right] \tag{4}$$

where $\tilde{B}$ denotes the bottom quantum number (distinguished from baryon number B). Note that for down-type quarks d, s, b, the corresponding flavor numbers $D, S, \tilde{B}$ are defined as $-(n_q - n_{\bar{q}})$, so that $I_3 = \frac{1}{2}(U + D)$, where $D = -(n_d - n_{\bar{d}})$; the formula above is self-consistent.

2.1.2 Restricted 3-quark approximation (u, d, s only)

$$\frac{Q}{e} = I_3 + \frac{1}{2}(B + S) \tag{5}$$

2.1.3 Proof via fundamental quark charges

Quark flavor	Q/e	B	$U + C + T + D + S + \tilde{B}$
$u/c/t$	$+2/3$	$1/3$	$+1$
$d/s/b$	$-1/3$	$1/3$	-1

The formula holds for all elementary quarks and all composite hadrons.

2.2 Definitions of Core Quantum Numbers

1. **Electric charge** Q: Charge of the particle, in units of the electron charge e.
2. **Third component of isospin** I_3:
   - Total isospin: $I = \frac{n-1}{2}$, where n is the number of particles with nearly identical mass
   - Allowed values: $I_3 = -I,\ -I+1,\ \ldots,\ I-1,\ I$
   - Relation to up/down quark number: $I_3 = \frac{1}{2}(U + D)$, where $D = -(n_d - n_{\bar{d}})$
3. **Hypercharge** Y: Sum of baryon number and all flavor quantum numbers.
4. **Baryon number** B:
$$B = \frac{1}{3}\left(n_q - n_{\bar{q}}\right) \tag{6}$$
where n_q = number of quarks, $n_{\bar{q}}$ = number of antiquarks.

5. **Flavor quantum numbers**:

   - Up number: $U = n_u - n_{\bar{u}}$
   - Down number: $D = -(n_d - n_{\bar{d}})$
   - Strangeness: $S = -(n_s - n_{\bar{s}})$
     - 1 strange quark s: $S = -1$; 1 anti-strange quark $\bar{s}$: $S = +1$
     - u, d, c, b, t quarks carry $S = 0$
   - Charm: $C = n_c - n_{\bar{c}}$
   - Bottom: $\tilde{B} = -(n_b - n_{\bar{b}})$
   - Top: $T = n_t - n_{\bar{t}}$

6. **Lepton number** L_l:

$$L_l = n_l - n_{\bar{l}} + n_{\nu_l} - n_{\bar{\nu}_l}, \quad l = e, \mu, \tau \tag{7}$$

   The total lepton number $L = L_e + L_\mu + L_\tau$ is conserved in all Standard Model interactions, while individual lepton flavors (L_e, L_μ, L_τ) are not conserved due to neutrino oscillations (which require neutrino masses and PMNS mixing).

2.3 Isospin Values for Quarks

Quark	u	d	$\bar{u}$	$\bar{d}$	$c, s, t, b, \bar{c}, \bar{s}, \bar{t}, \bar{b}$
I	$1/2$	$1/2$	$1/2$	$1/2$	0
I_3	$+1/2$	$-1/2$	$-1/2$	$+1/2$	0

Gauge bosons and the Higgs boson carry zero weak isospin. For leptons, left-handed leptons form $SU(2)_L$ doublets with $I = 1/2$ (e.g. $\begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix}$ with $I_3 = +1/2$ for ν_L and $I_3 = -1/2$ for e_L), while right-handed leptons are singlets with $I = 0$.

2.4 Worked Examples

2.4.1 Example 1: Ω^- (sss)

$$B = 1,\ S = -3 \implies Y = B + S = -2, \quad n = 1 \implies I = 0 \implies \frac{Q}{e} = -1 \tag{8}$$

2.4.2 Example 2: Δ^{++} (uuu)

$$B = 1,\ S = 0 \implies Y = B + S = 1 \tag{9}$$

For the isospin multiplet $\Delta^-, \Delta^0, \Delta^+, \Delta^{++}$, $n = 4$:

$$I = \frac{n-1}{2} = \frac{3}{2} \implies I_3 = \frac{3}{2} \quad (\text{for } \Delta^{++}) \implies \frac{Q}{e} = 2 \tag{10}$$

3 Relativistic Field Equations (by Spin)

Natural units $\hbar = c = k_B = 1$ are used by default; non-natural units are noted explicitly.

3.1 Spin-0: Klein–Gordon Equation (Scalar Bosons)

Energy-momentum dispersion relation:

$$E^2 = \mathbf{p}^2 c^2 + m^2 c^4 \tag{11}$$

Quantum operator substitution: $E \to i\hbar \frac{\partial}{\partial t},\ \mathbf{p} \to -i\hbar \nabla$

Covariant form:

$$\begin{cases} \left(\Box + \frac{m^2 c^2}{\hbar^2}\right) \psi = 0 \\ \Box = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \end{cases} \tag{12}$$

Natural unit simplification:

$$\left(\Box + m^2\right) \phi = 0 \tag{13}$$

3.2 Spin-1/2: Dirac Equation (Fermionic Spinors)

3.2.1 Time-dependent form

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-i\hbar c \alpha \cdot \nabla + \beta m c^2\right) \psi \tag{14}$$

Anticommutator identities:

- Anticommutator definition: $\{A, B\} = AB + BA$
- $\{\alpha^i, \alpha^j\} = 2\delta^{ij} I,\ \{\alpha^i, \beta\} = 0,\ \beta^2 = I$

3.2.2 Covariant form (natural units)

$$(i\gamma^\mu \partial_\mu - m)\, \psi = 0, \quad \not{\partial} = \gamma^\mu \partial_\mu \tag{15}$$

with $\gamma^0 = \beta,\ \gamma^i = \beta \alpha^i$ (Dirac gamma matrices).

3.2.3 Standard Dirac matrix representation

$$\gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{16}$$

3.2.4 Pauli matrices

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{17}$$

3.2.5 Chiral structure

- Chiral matrix (governs the $V - A$ structure of weak interactions):

$$\gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3 \tag{18}$$

- Chiral projection operators (project onto left/right-handed spinors):

$$P_{L,R} = \frac{1 \mp \gamma^5}{2} \tag{19}$$

3.2.6 Massless Weyl fermions

$$\begin{cases} \text{Right-handed: } \bar{\sigma}^\mu \partial_\mu \psi_R = 0 \\ \text{Left-handed: } \sigma^\mu \partial_\mu \psi_L = 0 \end{cases} \tag{20}$$

where $\sigma^\mu = (1, \sigma)$, $\bar{\sigma}^\mu = (1, -\sigma)$.

3.3 Spin-1: Vector Field Equations

3.3.1 Abelian (massless photon, QED): Maxwell equations

$$\begin{cases} \partial_\mu F^{\mu\nu} = j^\nu \\ \partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0 \end{cases} \tag{21}$$

3.3.2 Non-Abelian (gluons, W/Z bosons): Yang–Mills equations

$$\begin{cases} D_\mu F_a^{\mu\nu} = 0 \\ D_\mu F_{a\nu\rho} + D_\nu F_{a\rho\mu} + D_\rho F_{a\mu\nu} = 0 \end{cases} \tag{22}$$

3.4 Spin-$3/2$: Rarita–Schwinger Equation

$$\begin{cases} (i \not{\partial} - m)\psi^\mu = 0 \\ \gamma^\mu \psi_\mu = \partial_\mu \psi^\mu = 0 \end{cases} \tag{23}$$

3.5 Spin-2: Vacuum Einstein Field Equation (Gravity)

$$R_{\mu\nu} = 0 \tag{24}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor. The intrinsic parity of the massless graviton is $+1$.

4 Perturbative Calculation Framework

4.1 Fermi's Golden Rule

Computes the transition rate for an interaction process:

$$\Gamma_{fi} = \frac{2\pi}{\hbar} |T_{fi}|^2 \rho(E_f) \tag{25}$$

Definitions:

- Γ_{fi}: Number of transitions per unit time from initial state $|i\rangle$ to final state $|f\rangle$. The lifetime τ satisfies $\Gamma = 1/\tau$.
- T_{fi}: Transition matrix element, $T_{fi} = \langle f|\hat{H}_I|i\rangle$
- $|T_{fi}|^2$: Squared probability amplitude for the process
- $\rho(E_f)$: Density of final states, counting available phase-space states
- T_{fi} can be recast as the relativistically invariant amplitude $\mathcal{M}_{fi}$, calculated via Feynman Rules.

4.2 Feynman Path Integral Formalism

$$\langle x_f|e^{-iHt/\hbar}|x_i\rangle = \int \mathcal{D}x(t)\ e^{\frac{i}{\hbar}S[x(t)]} \tag{26}$$

Definitions:

- $\langle x_f|e^{-iHt/\hbar}|x_i\rangle$: Quantum transition amplitude
- $\int \mathcal{D}x(t)$: Functional integral over all paths connecting x_i and x_f
- $S[x(t)] = \int_0^t L(x, \dot{x})dt$: Classical action functional
- $e^{iS/\hbar}$: Phase weight factor for each path
- Hamiltonian: $H = \frac{\mathbf{p}^2}{2m} + V(x)$

4.3 Feynman Rules

4.3.1 Position-space ϕ^4 Theory

- Each propagator:

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{-ip\cdot(x-y)} \tag{27}$$

- Each vertex: $(-i\lambda)\int d^4z$
- Each external leg: 1
- Divide the final result by the symmetry factor.

4.3.2 Momentum-space General Rules

- Enforce four-momentum conservation at every vertex.
- Integrate over all undetermined loop momenta:

$$\int \frac{d^4p}{(2\pi)^4} \tag{28}$$

- Divide the result by the symmetry factor.

4.3.3 External Line Factors (asymptotic states)

1. **Spin-0 (scalar particles)**: Factor = 1
2. **Spin-**$1/2$ **(spinor particles)**
   - p: four-momentum, s: spin index, u, v: spinor wavefunctions
   - Incoming electron: $u^{(s)}(p)$
   - Outgoing electron: $\bar{u}^{(s)}(p) = u^\dagger\gamma^0$
   - Incoming positron: $\bar{v}^{(s)}(p)$
   - Outgoing positron: $v^{(s)}(p)$
3. **Spin-1 (vector particles)**
   - λ: polarization index, ε: polarization four-vector
   - Incoming photon: $\varepsilon_\mu^{(\lambda)}(p)$
   - Outgoing photon: $\varepsilon_\mu^{(\lambda)*}(p)$

4.3.4 Internal Line Propagators (virtual particles)

- Scalar particle (mass m):

$$\frac{i}{p^2 - m^2 + i\varepsilon} \tag{29}$$

- Spinor fermion (mass m):

$$\frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}, \quad \not{p} = \gamma^\mu p_\mu \tag{30}$$

- Massless photon:

$$\frac{-ig_{\mu\nu}}{p^2 + i\varepsilon} \tag{31}$$

- Massive vector boson (W, Z):

$$\frac{-i\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2}\right)}{p^2 - m^2 + i\varepsilon} \tag{32}$$

  High-energy approximation: $\dfrac{-ig_{\mu\nu}}{p^2 - m^2}$

- The $i\varepsilon$ prescription enforces causality; it is usually omitted in cross-section calculations.

4.3.5 Vertex Factors

- ϕ^4 theory: $-i\lambda$
- QED electron-photon vertex: $-ie\gamma^\mu$
- QCD quark-gluon vertex: $-ig_s T^a \gamma^\mu$
  - T^a: $SU(3)$ fundamental generators; g_s: strong coupling constant
- Weak interaction vertices
  - W-fermion vertex: $-i\dfrac{g_W}{\sqrt{2}}\gamma^\mu \dfrac{(1-\gamma^5)}{2}$
  - Z-fermion vertex: $-i\dfrac{g_W}{\cos\theta_W}\gamma^\mu \dfrac{(c_V - c_A\gamma^5)}{2}$

  where $c_V = T_3 - 2Q\sin^2\theta_W$ and $c_A = T_3$ depend on the specific fermion.

4.3.6 Supplementary Rules

- **Fermion lines**: Arrows indicate the direction of particle number flow; forward = particle, backward = antiparticle.
  - Closed fermion loops acquire a global factor of -1.
- **Relativistic normalization**: Additional factors from external wavefunction normalizations.
- **Symmetry factors**: For diagrams with identical particles, divide by a symmetry factor (e.g. 1/2) to avoid overcounting.
- **Loop diagrams**: Integrate over the undetermined loop momentum k:

$$\int \frac{d^4k}{(2\pi)^4} \tag{33}$$

4.3.7 Example: Muon-electron scattering

Process: $e^-(p_1)+\mu^-(p_2) \to e^-(p_3)+\mu^-(p_4)$, with momentum transfer $q = p_1 - p_3 = p_4 - p_2$

$$i\mathcal{M} = [\bar{u}(p_3)(-ie\gamma^\mu)u(p_1)] \cdot \left(\frac{-ig_{\mu\nu}}{q^2}\right) [\bar{u}(p_4)(-ie\gamma^\nu)u(p_2)] \tag{34}$$

- $\mathcal{M}$: Lorentz-invariant matrix element (equivalent to T in Fermi's Golden Rule)
- γ^μ: Dirac gamma matrix

5 Standard Model Gauge Group Structure

5.1 Conservation Laws Summary for All Interactions

Before delving into the individual gauge groups and their associated interactions, it is useful to have a global picture of which quantum numbers are conserved by which force. The table below summarizes the conservation laws for the electromagnetic, strong, and weak interactions.

Quantum Number	Electromagnetic	Strong	Weak
Electric Charge Q	✓	✓	✓
Baryon Number B	✓	✓	✓
Total Lepton Number L	✓	✓	✓
Color Charge	✓	✓	✓
Flavor Quantum Number	✓	✓	× (CKM/PMNS mixing)
Charge Conjugation C	✓	✓	×
Parity P	✓	✓	×
Time Reversal T	✓	✓	×
Combined CPT	✓	✓	✓

Note: Individual lepton flavors (L_e, L_μ, L_τ) are not conserved in weak interactions due to neutrino oscillations, but total lepton number $L = L_e + L_\mu + L_\tau$ remains conserved.

5.2 Full Standard Model Gauge Symmetry

$$G_{SM} = \underbrace{SU(3)_c}_{\text{Strong}} \times \underbrace{SU(2)_L \times U(1)_Y}_{\text{Electroweak}} \tag{35}$$

5.3 $SU(3)_c$ – Strong Interaction (QCD)

- c: Color charge quantum number
- **Role**: Describes quantum chromodynamics, the theory of strong interactions
- **Matter content**:
  - Quarks carry 3 color charges: red, green, blue
  - 8 massless gluons are the gauge bosons
- **Key features**:
  - Color confinement: Isolated free quarks cannot exist in nature
  - Asymptotic freedom: Interaction strength weakens at high energy / short distance
  - Leptons are $SU(3)_c$ color singlets (colorless) and do not feel the strong force
- **Mathematical definition**: Special Unitary group of degree 3; all 3×3 unitary matrices with determinant 1
- **Generators**: $T^a = \lambda^a/2,\ a = 1, \ldots, 8$
  - λ^a: Gell-Mann matrices; $8 = 3^2 - 1$ independent generators
- **Irreducible representations**: **3** (quarks), **3** (antiquarks), **8** (gluons)

5.4 $SU(2)_L$ – Weak Interaction (Left-Handed Chiral Sector)

- L: Left-handed chiral projection
- **Role**: Governs charged weak interactions; only couples to left-handed fermions
- **Matter content**:
  - Left-handed fermions form $SU(2)_L$ doublets, e.g.

$$\begin{pmatrix} \nu_e \\ e_L \end{pmatrix}, \quad \begin{pmatrix} u_L \\ d_L \end{pmatrix} \tag{36}$$

  - Right-handed fermions are $SU(2)_L$ singlets and have no charged weak coupling
- **Key feature**: The chiral nature is the origin of parity violation in weak interactions
- **Gauge bosons**: W^1, W^2, W^3. After electroweak symmetry breaking, these become massive $W^\pm$ and mix with the $U(1)_Y$ B boson to form the neutral Z^0.
- **Mathematical definition**: Special Unitary group of degree 2; all 2×2 unitary matrices with determinant 1
- **Generators**: $T^a = \sigma^a/2,\ a = 1, 2, 3$; $3 = 2^2 - 1$ generators
- **Lie algebra**:

$$[T^a, T^b] = i\varepsilon^{abc} T^c \tag{37}$$

  where ε^{abc} is the fully antisymmetric Levi-Civita symbol.
- **Representations**:
  - **2** (fundamental): Left-handed fermion doublets
  - **1** (singlet): Right-handed fermions
  - **3** (adjoint): W boson weak isospin triplet

5.5 $U(1)_Y$ – Hypercharge Gauge Symmetry

- Y: Weak hypercharge quantum number
- **Role**: Part of electroweak unification
- **Gauge boson**: B_μ
- **Relation to electric charge**: Gell-Mann–Nishijima formula

$$Q = I_3 + \frac{Y}{2} \tag{38}$$

- **Mathematical definition**: Degree-1 unitary group; all 1×1 unitary phase factors $e^{i\theta}$
- Single generator: 1; trivial abelian Lie algebra
- **Representation**: Hypercharge Y; field transformation: $\psi \to e^{iY\theta}\psi$

5.6 Electroweak Symmetry Breaking

$$SU(2)_L \times U(1)_Y \xrightarrow{\text{symmetry breaking}} U(1)_{EM} \tag{39}$$

The original gauge multiplet (W^1, W^2, W^3, B) mixes into physical electroweak bosons $(W^\pm, Z^0, \gamma)$ via the Weinberg angle θ_W:

$$\begin{cases} W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2) \\ \gamma = \sin\theta_W W^3 + \cos\theta_W B \\ Z^0 = \cos\theta_W W^3 - \sin\theta_W B \end{cases} \tag{40}$$

6 Full Standard Model Lagrangian

$$\mathcal{L}_{SM} = \mathcal{L}_{gauge} + \mathcal{L}_{fermion} + \mathcal{L}_{Higgs} + \mathcal{L}_{Yukawa} \tag{41}$$

6.1 Gauge Field (Yang–Mills) Sector

$$\mathcal{L}_{gauge} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} \tag{42}$$

Field strength tensors:

1. Gluon ($SU(3)_c$, non-abelian, self-interacting):

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu \tag{43}$$

2. Weak W boson ($SU(2)_L$, non-abelian, self-interacting):

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + g \varepsilon^{ijk} W^j_\mu W^k_\nu \tag{44}$$

3. Hypercharge B boson ($U(1)_Y$, abelian, no self-interactions):

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{45}$$

6.2 Fermion Kinetic (Matter) Sector

$$\mathcal{L}_{fermion} = \sum_{fermions} i \bar{\psi} \gamma^\mu D_\mu \psi \tag{46}$$

Covariant derivative (includes all three gauge interactions):

$$D_\mu = \partial_\mu - i g_s G^a_\mu T^a - i g W^i_\mu t^i - i g' \frac{Y}{2} B_\mu \tag{47}$$

- γ^μ: Dirac matrices
- g_s, g, g': Coupling constants for strong, weak, and hypercharge interactions
- T^a: $SU(3)_c$ generators
- $t^i = \sigma^i/2$: $SU(2)_L$ generators
- Y: $U(1)_Y$ hypercharge operator

6.3 Higgs Sector

$$\mathcal{L}_{Higgs} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi) \tag{48}$$

- Higgs field Φ: $SU(2)_L$ complex doublet with hypercharge $Y = +1$

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{49}$$

(The hypercharge assignment $Y = +1$ follows from $Q = I_3 + Y/2$ and the fact that the neutral component ϕ^0 has $Q = 0$ and $I_3 = -1/2$.)

- Higgs covariant derivative:

$$D_\mu \Phi = \left(\partial_\mu - igW_\mu^i \frac{\sigma^i}{2} - ig' \frac{Y}{2} B_\mu \right) \Phi \tag{50}$$

- Higgs potential (drives spontaneous symmetry breaking):

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \quad (\mu^2 > 0,\ \lambda > 0) \tag{51}$$

  The vacuum expectation value is $v = \sqrt{\mu^2/\lambda} \approx 246$ GeV.

6.4 Yukawa Coupling (Fermion Mass Generation)

$$\mathcal{L}_{Yukawa} = -y_e \bar{L}_e \Phi e_R - y_d \bar{Q}_L \Phi d_R - y_u \bar{Q}_L \tilde{\Phi} u_R + \text{h.c.} \tag{52}$$

Charge-conjugated Higgs doublet: $\tilde{\Phi} = i\sigma^2 \Phi^*$
Fermion masses after electroweak symmetry breaking (VEV v):

$$m_e = y_e \frac{v}{\sqrt{2}}, \quad m_u = y_u \frac{v}{\sqrt{2}}, \quad m_d = y_d \frac{v}{\sqrt{2}} \tag{53}$$

7 Quantum Electrodynamics (QED) – Electromagnetic Interaction

7.1 QED Action and Lagrangian Density

Action definition:

$$S = \int d^4x \; \mathcal{L}(\phi, \partial_\mu \phi) \tag{54}$$

Euler–Lagrange equation of motion:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \tag{55}$$

7.1.1 Component Lagrangians

1. Klein–Gordon scalar (spin-0):

$$\mathcal{L}_{KG} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 \tag{56}$$

2. Dirac fermion (spin-1/2):

$$\mathcal{L}_{Dirac} = \bar{\psi} \left(i\gamma^\mu \partial_\mu - m \right) \psi \tag{57}$$

3. Maxwell photon (spin-1):

$$\mathcal{L}_{Maxwell} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \tag{58}$$

   Maxwell field equations: $\partial_\mu F^{\mu\nu} = j^\nu$

7.1.2 Full QED Lagrangian

$$\mathcal{L}_{QED} = \underbrace{i\bar{\psi}\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi}_{\text{Free Dirac electron}} - \underbrace{\frac{1}{4} F_{\mu\nu} F^{\mu\nu}}_{\text{Free Maxwell photon}} - \underbrace{e(\bar{\psi}\gamma^\mu \psi) A_\mu}_{e-\gamma \text{ local invariant interaction}} \tag{59}$$

Supporting identities:

- Dirac adjoint: $\bar{\psi} = \psi^\dagger \gamma^0$
- QED covariant derivative: $D_\mu = \partial_\mu + ieA_\mu$
- Field strength tensor: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

7.2 Gauge Invariance

$U(1)$ gauge transformation operator: $U = e^{-i\theta}$

1. **Global gauge transformation** (θ = constant, spacetime-independent):

$$\psi \to \tilde{\psi} = e^{-i\theta} \psi \tag{60}$$

   $\mathcal{L}_{QED}$ remains invariant.

2. **Local gauge transformation** ($\theta = \lambda(x)$, spacetime-dependent):

$$\psi \to \tilde{\psi} = e^{-ie\lambda(x)} \psi, \quad D_\mu \tilde{\psi} = e^{-ie\lambda} D_\mu \psi \tag{61}$$

   Gauge field shift: $A_\mu \to \tilde{A}_\mu = A_\mu + \partial_\mu \lambda$
   The full QED Lagrangian remains invariant under local $U(1)$ transformations.

7.3 QED Vertex and Feynman Rules

The QED interaction vertex is:

$$-ie\gamma^{\mu} \tag{62}$$

For a generic QED process, the amplitude is constructed by:

- Assigning fermion propagators: $\dfrac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$
- Assigning photon propagators: $\dfrac{-ig_{\mu\nu}}{p^2 + i\varepsilon}$
- Multiplying by vertex factors $-ie\gamma^{\mu}$ at each interaction point
- Enforcing momentum conservation at each vertex

8 Quantum Chromodynamics (QCD) – Strong Interaction

8.1 Color Charge and Gluon States

- Quark colors: Red R, Green G, Blue B
- Antiquark colors: Anti-red $\bar{R}$, Anti-green $\bar{G}$, Anti-blue $\bar{B}$

Color tensor product decomposition:

$$3 \otimes \bar{3} = \mathbf{8} \oplus \mathbf{1} \tag{63}$$

8.1.1 8 physical gluon octet states

1. 6 off-diagonal (edge) states: $R\bar{G}, G\bar{R}, R\bar{B}, B\bar{R}, G\bar{B}, B\bar{G}$
2. 2 traceless diagonal (center) states:

$$\frac{1}{\sqrt{2}}(R\bar{R} - G\bar{G}), \quad \frac{1}{\sqrt{6}}(R\bar{R} + G\bar{G} - 2B\bar{B}) \tag{64}$$

8.1.2 1 unobservable color singlet

$$\frac{1}{\sqrt{3}}(R\bar{R} + G\bar{G} + B\bar{B}) \tag{65}$$

Colorless; does not participate in strong interactions.

8.2 Color Confinement

All observable hadrons are color singlets (color-neutral):

- Baryons: $R + G + B$
- Antibaryons: $\bar{R} + \bar{G} + \bar{B}$
- Mesons: $R\bar{R}$ or $G\bar{G}$ or $B\bar{B}$

8.3 Asymptotic Freedom

At high energy / short inter-quark distance, the strong coupling constant α_s becomes small, permitting perturbative calculations.

The running of the strong coupling is governed by the QCD β-function:

$$\beta(\alpha_s) = \mu^2 \frac{d\alpha_s}{d\mu^2} = -\frac{11N_c - 2N_f}{12\pi}\alpha_s^2 + \mathcal{O}(\alpha_s^3) \tag{66}$$

For $N_c = 3$ and $N_f \leq 16$, the first coefficient is negative, leading to asymptotic freedom.

8.3.1 QCD static quark potential

$$V_{\text{QCD}}(r) = -\frac{4}{3}\frac{\alpha_s}{r} + kr \tag{67}$$

Parameters:

- Strong coupling: $\alpha_s \approx 0.3$
- String tension: $k = 1$ GeV/fm
- Short distance: Coulomb-like term dominates (asymptotic freedom)
- Long distance: Linear confining term dominates (color confinement)

8.4 QCD Vertex and Feynman Rules

The QCD quark-gluon vertex is:

$$-ig_s T^a \gamma^\mu \tag{68}$$

where T^a are the $SU(3)$ generators ($a = 1, \ldots, 8$).

QCD also features:

- Gluon self-interactions (3-gluon and 4-gluon vertices) due to the non-abelian nature of $SU(3)$.
- Ghost fields in covariant gauges to maintain unitarity.

8.5 Experimental Evidence: R Ratio in e^+e^- Annihilation

Definition: Ratio of total hadronic cross section to muon pair cross section

$$R = \frac{\sigma_{\text{hadrons}}}{\sigma_{\mu^+\mu^-}} = \sum_{\text{active quarks}} Q_q^2 \tag{69}$$

With 3 color degrees of freedom per quark:

$$R = 3\sum Q_q^2 \tag{70}$$

Numerical predictions at different energy thresholds:

- u, d, s active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = 2 \tag{71}$$

- u, d, s, c active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2\right] = \frac{10}{3} \tag{72}$$

- u, d, s, c, b active:

$$R = 3\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = \frac{11}{3} \tag{73}$$

Center-of-mass energy: $\sqrt{s} = E_{e^-} + E_{e^+}$
Experimental measurements match the factor-of-3 color enhancement, confirming 3 color charges per quark.

8.6 Quark Model: Baryon Flavor Wavefunctions

Total baryon wavefunction factorization:

$$\Psi = \Psi(\text{space})\,\Psi(\text{spin})\,\Psi(\text{flavor})\,\Psi(\text{color}) \tag{74}$$

8.6.1 $SU(3)_f$ Three-Quark Decomposition

$$3 \otimes 3 \otimes 3 = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1} \tag{75}$$

Symmetry classes:

- S Fully symmetric: **10** decuplet; $\Psi(a,b,c) = \Psi(b,a,c)$ for all permutations
- A Fully antisymmetric: **1** singlet; $\Psi(a,b,c) = -\Psi(b,a,c)$ for all swaps
- M Mixed symmetry: two **8** octets; antisymmetric under specific quark pair swaps

8.6.2 Fully Symmetric Decuplet (10)

Normalized basis states:

- Pure-flavor states: $ddd,\ uuu,\ sss$
- Two-identical-flavor states:

$$\frac{1}{\sqrt{3}}(dds + dsd + sdd), \quad \frac{1}{\sqrt{3}}(uus + usu + suu) \tag{76}$$

$$\frac{1}{\sqrt{3}}(udd + udu + duu), \quad \frac{1}{\sqrt{3}}(dss + sds + ssd) \tag{77}$$

$$\frac{1}{\sqrt{3}}(uss + sus + ssu) \tag{78}$$

- Three-different-flavor symmetric state:

$$\frac{1}{\sqrt{6}}(uds + usd + dus + dsu + sud + sdu) \tag{79}$$

8.6.3 Fully Antisymmetric Singlet (1)

$$\frac{1}{\sqrt{6}}\big(uds - usd + sud - sdu + dsu - dus\big) \tag{80}$$

8.6.4 Mixed-Symmetry Octets $(\mathbf{8},\mathbf{8})$

1. Ψ_{12}: Antisymmetric under quark $1 \leftrightarrow 2$ exchange
   Building blocks: $\dfrac{ud-du}{\sqrt{2}},\ \dfrac{ds-sd}{\sqrt{2}},\ \dfrac{us-su}{\sqrt{2}}$
   Linear combinations:

$$\frac{1}{2}\big[(us-su)d + (ds-sd)u\big], \quad \frac{1}{\sqrt{12}}\big[2(ud-du)s + (us-su)d - (ds-sd)u\big] \tag{81}$$

2. Ψ_{23}: Antisymmetric under quark $2 \leftrightarrow 3$ exchange
   Building blocks: $d\dfrac{ud-du}{\sqrt{2}},\ d\dfrac{ds-sd}{\sqrt{2}},\ u\dfrac{ud-du}{\sqrt{2}},\ u\dfrac{us-su}{\sqrt{2}},\ s\dfrac{ds-sd}{\sqrt{2}},\ s\dfrac{us-su}{\sqrt{2}}$
   Linear combinations:

$$\frac{1}{2}\big[d(us-su)+u(ds-sd)\big], \quad \frac{1}{\sqrt{12}}\big[2s(ud-du)+d(us-su)-u(ds-sd)\big] \tag{82}$$

Linear dependence relation:

$$\Psi_{13} = \Psi_{12} + \Psi_{23} \tag{83}$$

Ψ_{13} is not an independent basis state.

8.6.5 Clebsch–Gordan Coefficients

Angular momentum coupling expansion:

$$|j_1 m_1, j_2 m_2\rangle = \sum_{\substack{j \\ m=m_1+m_2}} |jm\rangle\langle jm|j_1 m_1, j_2 m_2\rangle \tag{84}$$

Allowed ranges:

- Total angular momentum: $j = |j_1 - j_2|,\ |j_1 - j_2|+1,\ \ldots,\ j_1 + j_2$
- Magnetic projection: $m = m_1 + m_2$
- Individual projections: $m_1 = -j_1, \ldots, j_1$; $m_2 = -j_2, \ldots, j_2$; $m = -j, \ldots, j$

9 Weak Interaction

The weak interaction is described by the $SU(2)_L \times U(1)_Y$ sector of the Standard Model. It is characterized by:

- **Chirality**: Only left-handed fermions participate in charged current interactions.
- **Parity violation**: The $V - A$ structure of weak currents maximally violates parity.
- **Massive mediators**: $W^{\pm}$ and Z^0 bosons are massive due to electroweak symmetry breaking.

9.1 Charged Current Interaction

The charged current interaction couples left-handed fermion doublets via $W^{\pm}$ bosons.

For a generic fermion doublet (f_1, f_2), the charged current Lagrangian is:

$$\mathcal{L}_{CC} = -\frac{g}{\sqrt{2}} \bar{f}_1 \gamma^\mu P_L f_2 W_\mu^+ + \text{h.c.} \tag{85}$$

where $P_L = (1 - \gamma^5)/2$ is the left-handed projection operator.

This mediates processes such as:

- β-decay: $d \to u + e^- + \bar{\nu}_e$
- Muon decay: $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$
- Neutrino scattering: $\nu_\mu + e^- \to \mu^- + \nu_e$

9.2 Neutral Current Interaction

The neutral current interaction is mediated by the Z^0 boson:

$$\mathcal{L}_{NC} = -\frac{g}{2\cos\theta_W} \bar{\psi} \gamma^\mu (c_V - c_A \gamma^5) \psi Z_\mu \tag{86}$$

where:

$$c_V = T_3 - 2Q \sin^2\theta_W, \qquad c_A = T_3 \tag{87}$$

depend on the specific fermion.

Key properties:

- The Z^0 couples to both left- and right-handed fermions (unlike the $W^{\pm}$).
- Neutral currents do not change fermion flavor (e.g., $Z^0 \to e^+e^-$ but not $Z^0 \to e^{\pm}\mu^{\mp}$).

9.3 Weak Interaction Vertex Factors

- W-fermion vertex: $-i\frac{g_W}{\sqrt{2}} \gamma^\mu \frac{(1 - \gamma^5)}{2}$
- Z-fermion vertex: $-i\frac{g_W}{\cos\theta_W} \gamma^\mu \frac{(c_V - c_A \gamma^5)}{2}$

9.4 Flavor Mixing in Weak Interactions

The weak interaction eigenstates are not the same as the mass eigenstates for quarks and neutrinos.

9.4.1 Quark Mixing: CKM Matrix

Up-type weak eigenstates match mass eigenstates exactly:

$$\begin{pmatrix} u' \\ c' \\ t' \end{pmatrix} = \begin{pmatrix} u \\ c \\ t \end{pmatrix} \tag{88}$$

Down-type weak eigenstates mix via the Cabibbo–Kobayashi–Maskawa matrix V_{CKM}:

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = V_{CKM} \begin{pmatrix} d \\ s \\ b \end{pmatrix} \tag{89}$$

Wolfenstein parametrization:

$$V_{CKM} = \begin{pmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} \tag{90}$$

Parameters: $A \approx 0.83,\ \rho \approx 0.13,\ \eta \approx 0.35,\ \lambda = \sin\theta_C = 0.225.$

9.4.2 Lepton Mixing: PMNS Matrix

Charged lepton weak eigenstates equal mass eigenstates:

$$\begin{pmatrix} e' \\ \mu' \\ \tau' \end{pmatrix} = \begin{pmatrix} e \\ \mu \\ \tau \end{pmatrix} \tag{91}$$

Neutrino flavor $\leftrightarrow$ mass eigenstate mixing via the Pontecorvo–Maki–Nakagawa–Sakata matrix U_{PMNS}:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{PMNS} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \tag{92}$$

Standard factorized parametrization:

$$U_{PMNS} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{i\alpha_1/2} & 0 & 0 \\ 0 & e^{i\alpha_2/2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{93}$$

where $c_{ij} = \cos\theta_{ij},\ s_{ij} = \sin\theta_{ij}$, δ is the Dirac CP-violating phase, and α_1, α_2 are Majorana phases.

Measured mixing angles:

$$\theta_{12} \approx 34^\circ, \quad \theta_{23} \approx 45^\circ, \quad \theta_{13} \approx 9^\circ \tag{94}$$

9.5 Neutrino Physics

9.5.1 Helicity

$$h = \frac{\mathbf{S} \cdot \hat{P}}{|\mathbf{S}|} \tag{95}$$

- $\mathbf{S}$: Spin vector; $\hat{P}$: Unit momentum vector
- $h < 0$: Left-handed (LH), spin antiparallel to momentum
- $h > 0$: Right-handed (RH), spin parallel to momentum

Standard Model chiral rules:

- Photons are produced in equal LH and RH helicity states
- All active neutrinos are left-handed: ν_L
- All active antineutrinos are right-handed: $\bar{\nu}_R$

CP transformation of neutrino helicity:

$$\nu_L \xrightarrow{\hat{C}} \bar{\nu}_L \xrightarrow{\hat{P}} \bar{\nu}_R \tag{96}$$

$\bar{\nu}_L$ does not exist in nature; massless neutrinos violate CP intrinsically.

9.5.2 Neutrino Detection

Charged-current detection reactions:

$$\nu_e + n \to p + e^-, \quad \nu_\mu + n \to p + \mu^- \tag{97}$$

Typical mean free path in matter: $d_p \sim 1.5 \times 10^{16}$ m

9.5.3 Neutrino Sources

1. Accelerator neutrino beams
2. Solar proton-proton fusion chain:

$$p + p \to d + e^+ + \nu_e \tag{98}$$
$$d + p \to {}^3\mathrm{He} + \gamma \tag{99}$$
$${}^3\mathrm{He} + {}^3\mathrm{He} \to {}^4\mathrm{He} + p + p \tag{100}$$

3. Atmospheric neutrinos (cosmic ray air showers)
4. Reactor electron antineutrinos (nuclear beta decay):

$$(Z, A) \to (Z + 1, A) + e^- + \bar{\nu}_e \tag{101}$$

where Z = proton number, A = mass number, N = neutron number.

9.5.4 Neutrino Oscillations

Oscillation requires non-zero neutrino mass: $m_\nu > 0$
Flavor states oscillate via PMNS mixing: $\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$

10 Discrete Symmetries and CP Violation

10.1 Charge Conjugation Operator $\hat{C}$

State transformation:

$$\hat{C}|\text{particle}\rangle = \eta_C|\text{antiparticle}\rangle \tag{102}$$

- For self-conjugate neutral particles (e.g. π^0), $\eta_C = \pm 1$ is an observable phase.
- For non-self-conjugate particles (e.g. e^-), $\hat{C}$ simply maps to the antiparticle state without an additional phase factor.

Transformation rules:

- Electric charge: $Q \to -Q$
- Baryon number: $B \to -B$
- Lepton number: $L \to -L$
- All flavor quantum numbers flip sign
- Electromagnetic fields: $\mathbf{E} \to -\mathbf{E},\ \mathbf{B} \to -\mathbf{B}$
- Mass m, energy E, momentum $\mathbf{p}$, spin $\mathbf{S}$: Unchanged

10.2 Parity Operator $\hat{P}$

State transformation:

$$\hat{P}|\psi(\mathbf{x}, t)\rangle = \eta_P|\psi(-\mathbf{x}, t)\rangle \tag{103}$$

10.2.1 Intrinsic parity assignments

1. Fermions
   - Particle: $\eta_P = +1$
   - Antiparticle: $\eta_P = -1$
2. Bosons
   - Spin-0 pseudoscalar ($\pi^0, \pi^\pm$): $\eta_P = -1$
   - Spin-0 scalar (Higgs H): $\eta_P = +1$
   - Spin-1 vector ($\gamma, g, W^\pm, Z^0$): $\eta_P = -1$
   - Spin-2 graviton: $\eta_P = +1$

10.2.2 Hadron parity

$$\begin{cases} \eta_{P,\text{meson}} = (-1)^L \eta_{q_1}\eta_{\bar{q}_2} \\ \eta_{P,\text{baryon}} = (-1)^L \eta_{q_1}\eta_{q_2}\eta_{q_3} \end{cases} \tag{104}$$

where L is the total orbital angular momentum between constituent quarks.

10.3 Time Reversal Operator $\hat{T}$

State transformation:

$$\hat{T}|\psi(\mathbf{x},t)\rangle = \eta_T|\psi^*(\mathbf{x},-t)\rangle \tag{105}$$

Quantum number flips:

- Momentum: $\mathbf{P} \to -\mathbf{P}$
- Spin: $\mathbf{S} \to -\mathbf{S}$

10.4 CPT Theorem

Unitary condition:

$$(\hat{C}\hat{P}\hat{T})^\dagger(\hat{C}\hat{P}\hat{T}) = \mathbb{I} \tag{106}$$

For any quantum state:

$$\hat{C}\hat{P}\hat{T}|\Psi\rangle = e^{i\theta}|\Psi\rangle \tag{107}$$

CPT symmetry is exact in all local Lorentz-invariant quantum field theories.

10.5 CP Violation in the Neutral Kaon System

10.5.1 Flavor eigenstates

- $|K^0\rangle = |d\bar{s}\rangle,\ S = +1$
- $|\bar{K}^0\rangle = |s\bar{d}\rangle,\ S = -1$

10.5.2 Pure CP eigenstates (theoretical)

$$\begin{cases} |K_1\rangle = \dfrac{1}{\sqrt{2}}\big(|K^0\rangle + |\bar{K}^0\rangle\big), & CP = +1 \quad \text{(decays to } 2\pi\text{)} \\ |K_2\rangle = \dfrac{1}{\sqrt{2}}\big(|K^0\rangle - |\bar{K}^0\rangle\big), & CP = -1 \quad \text{(decays to } 3\pi\text{)} \end{cases} \tag{108}$$

10.5.3 Physical mass eigenstates

If CP were exact: $|K_S\rangle = |K_1\rangle$ (short-lived), $|K_L\rangle = |K_2\rangle$ (long-lived).

Experimentally, the rare decay $K_L \to \pi^+\pi^-$ is observed, proving CP violation. The physical long-lived state is a mixed state:

$$|K_L\rangle = \frac{1}{\sqrt{1+|\varepsilon|^2}}\big(|K_2\rangle + \varepsilon|K_1\rangle\big) \tag{109}$$

- $\varepsilon \approx 2.2 \times 10^{-3}$: CP violation parameter
- Semileptonic decay asymmetry:

$$\frac{\Gamma(K_L \to e^+\nu_e\pi^-)}{\Gamma(K_L \to e^-\bar{\nu}_e\pi^+)} \approx 1.00648 \tag{110}$$

By the CPT theorem, CP violation implies time reversal symmetry violation.

11 Beyond the Standard Model

11.1 Fundamental Coupling Constants and Interaction Ranges

Effective range generic formula: $R = \frac{\hbar}{mc}$ (Compton wavelength form)

Force	Fine-structure constant	Effective Range
Electromagnetic	$\alpha = \frac{e^2}{4\pi\hbar c} \approx \frac{1}{137}$	∞
Strong	$\alpha_s = \frac{g_s^2}{4\pi} \approx 1$	1.4×10^{-15} m
Weak	$\alpha_W = \frac{g^2}{4\pi} \approx \frac{1}{30}$	2.5×10^{-18} m
Gravitation	$\alpha_G = \frac{G_N m_p^2}{\hbar c} \approx 10^{-39}$	∞

G_N: Newtonian gravitational constant

11.2 Limitations of the Standard Model

1. Original formulation predicts massless neutrinos, conflicting with oscillation data
2. Many undetermined free parameters: fermion masses, coupling constants, mixing angles
3. Strong and electroweak forces are not unified (requires Grand Unified Theory, GUT)
4. No description of gravity (requires Theory of Everything, ToE)
5. Lepton anomalous magnetic moment g-factor: 3σ discrepancy between theory and experiment

11.2.1 Anomalous magnetic moment

Magnetic moment operator:

$$\mu_l = g_l \frac{e\hbar}{2mc}\mathbf{S} \tag{111}$$

Theoretical predictions:

- Non-relativistic Schrödinger: $g_l = 1$
- Relativistic Dirac: $g_l = 2$
- QFT radiative corrections: $g_l = 2(1 + a_l)$

Measured values:

- Electron: $a_e \approx 1.15965 \times 10^{-3}$
- Muon: $a_\mu \approx 1.16592 \times 10^{-3}$

11.3 Gauge Coupling Unification

The coupling constants of strong, weak, and electromagnetic forces converge to a common value at the grand unification scale $\sim 10^{16}$ GeV.

11.4 Dark Matter: WIMP Candidate

Weakly Interacting Massive Particle (WIMP):

- Mass: $M_{\text{WIMP}} \sim 100 \sim 200$ GeV/c^2
- Electrically neutral, cosmologically stable

11.5 Supersymmetry (SUSY)

Core symmetry: Maps Standard Model fermions $\leftrightarrow$ boson superpartners, and vice versa.

SM Particle (spin)	Superpartner (spin)	Name
Quark q (1/2)	Squark $\tilde{q}$ (0)	Squark
Electron e (1/2)	Selectron $\tilde{e}$ (0)	Selectron
Muon μ (1/2)	Smuon $\tilde{\mu}$ (0)	Smuon
Tau τ (1/2)	Stau $\tilde{\tau}$ (0)	Stau
Neutrino ν (1/2)	Sneutrino $\tilde{\nu}$ (0)	Sneutrino
W boson (1)	Wino $\tilde{W}$ (1/2)	Wino
Z boson (1)	Zino $\tilde{Z}$ (1/2)	Zino
Photon γ (1)	Photino $\tilde{\gamma}$ (1/2)	Photino
Gluon g (1)	Gluino $\tilde{g}$ (1/2)	Gluino

Neutral gaugino-higgsino mixing produces the neutralino, a leading WIMP dark matter candidate.

11.6 Seesaw Mechanism for Neutrino Masses

In the Standard Model, neutrino masses can be generated via the seesaw mechanism, which introduces heavy right-handed neutrinos N_R:

$$\mathcal{L}_{\text{seesaw}} = -y_\nu \bar{L}\tilde{\Phi}N_R - \frac{1}{2}M_N \bar{N}_R^c N_R + \text{h.c.} \tag{112}$$

For $M_N \gg y_\nu v$, the light neutrino mass is suppressed:

$$m_\nu \sim \frac{(y_\nu v)^2}{M_N} \tag{113}$$

This naturally explains the smallness of neutrino masses.

A Appendix 1: Lie Group Notation Reference

Symbol	Full Name	Definition
$GL(n)$	General Linear Group	All invertible $n \times n$ complex matrices
$SL(n)$	Special Linear Group	$GL(n)$ matrices with determinant = 1
$O(n)$	Orthogonal Group	Real matrices preserving real vector norm: $R^T R = I$
$SO(n)$	Special Orthogonal Group	$O(n)$ matrices with determinant = 1; describes rotations
$U(n)$	Unitary Group	Complex matrices preserving complex vector norm: $U^\dagger U = I$
$SU(n)$	Special Unitary Group	$U(n)$ matrices with determinant = 1
$Sp(2n)$	Symplectic Group	Matrices preserving symplectic bilinear form

Key letter mnemonics:

- S = Special (determinant 1)
- O = Orthogonal (real norm-preserving)
- U = Unitary (complex norm-preserving)

B Appendix 2: Matrix Definitions

- Unitary matrix U: $U^\dagger U = UU^\dagger = I$
- Symplectic matrix M: $M^T \Omega M = \Omega$, with symplectic metric

$$\Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \tag{114}$$

for $2n$-dimensional phase space.